



OPEN Numerical modeling and neural network optimization for advanced solar panel efficiency

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Maximizing output from renewable solar panels requires higher efficiency. Conventionally, such optimization techniques—MPPT (Maximum Power Point Tracking) along with heuristic algorithms—suffer significantly from slow adaptability and track sub optimality under dynamic environments. This article proposes a numerical modeling framework from hybrid AI models, combining physics-informed neural networks and RL for real-time optimization of orientation in solar panels. The methodology uses numerical modeling for precise energy transformation analysis, and deep learning-based optimization dynamically adjusts the angles of panels to maximize power output. A self-learning adaptive neural network is developed to improve tracking accuracy based on real-time irradiance and temperature variations. Moreover, an Edge AI architecture is introduced to make low-latency decisions with reduced dependency on cloud computation, thus improving the efficiency of the system. Besides, an advanced hybrid model based on CNN-LSTM is applied to solar energy forecasting for predictive control of the maximum energy yield. Experimental validation was performed using UTL 335W and 330W PV modules, where real-time data acquisition was followed by AI-driven optimization. Results show an increase in energy yield by 10–15% compared to traditional MPPT systems, while computations are performed 40–50% faster using AI-based numerical modeling. The proposed approach achieves 25% lower forecasting error (RMSE/MAE) and 30% reduced power consumption through Edge AI implementation. This study sets up a new paradigm for AI-integrated solar optimization, which ensures real-time adaptability and enhanced performance in practical deployment. The findings advance the intelligent solar tracking and set a new benchmark for AI-driven renewable energy management.

Keywords Numerical modeling, Neural network optimization, Solar panel efficiency, Physics-informed neural networks (PINNs), Reinforcement learning, Edge AI, CNN-LSTM, Solar forecasting

The global shift to renewable energy has made solar energy a leading sustainable energy generation component. As the world's reserve of fossil fuel is being rapidly depleted and fears of climate change are increasing, the technology for solar photovoltaic (PV) has captured much attention owing to its extensive availability and nontoxic character. Yet, despite all technological advancements in PV, one of the most lingering issues remains optimizing the efficiency of solar panels. The efficiency of solar panels generally depends on its orientation, environment, and capabilities to track or capture maximum sun irradiance along the day. The conventional type of solar tracking mechanism, be it single- or dual-axis tracking systems, have been significantly applied to raise the energy output but are severely dependent on either static or programmed algorithms that constrain them from any adaptability within real-time environment changes¹. The most used optimization algorithms include Traditional Maximum Power Point Tracking (MPPT), Particle Swarm Optimization (PSO), and Genetic Algorithms (GA). However, these techniques have some disadvantages inherent to them with their traditional methods of approach. Practically, most of the methods are heuristics or rule-based optimizations, failing to adapt when there are rapid changes in solar irradiance, temperature, and cloud cover occurring over a short time period. The algorithms also face some high computational overhead, slow convergence rates, and sensitivity to initial conditions. In such dynamic weather conditions, the energy harvesting becomes inefficient. Further, the MPPT techniques that are widely applied in solar PV systems have fixed voltage-current characteristics that do

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not depict real-time shading, dust deposition, or atmospheric disturbances. This limitation necessitates the development of intelligent, adaptive, and computationally efficient solar tracking solutions that can respond to environmental changes in real-time. In the last few years, AI/ML-based methods have emerged as promising solutions for real-time solar panel optimization². History and real-time environmental data-based approaches to predict optimal orientations for the solar panels, which enables higher energy yields with increased efficiency. Techniques that fall under deep learning models, such as ANNs and RL, were found to learn complex patterns in solar irradiance and temperature variability to adapt well to constantly changing environmental conditions. Most of the traditional AI models are not included with physics-based solar energy transformation equations that would sometimes provide answers that may not necessarily make for perfect physics or less than perfect. To overcome this, researchers have been looking into Physics-Informed Neural Networks (PINNs), integrating fundamental governing equations, such as thermodynamics, heat transfer, and photovoltaic electrical modeling, into the AI framework in a way that optimization decisions comply with real solar energy dynamics³. The solar tracking technologies have undergone tremendous evolution, beginning from the mechanical system tracking to algorithm-driven optimizations and further developments into AI-integrated intelligent control mechanisms. The first form of passive solar tracking was the thermal expansion materials that used temperature-induced expansion and contraction for panel orientation shift. These were less efficient as they did not work with much precision in fluctuating environmental conditions. Later, active tracking systems using motors, hydraulic actuators, and electronic sensors evolved to enhance the harvesting of solar energy by dynamically orienting the panels⁴. The mechanically controlled tracking systems improved energy capture but experienced high power consumption, frequent maintenance problems, and mechanical wear over time that limited the long-term efficiency and scalability of this approach. With improvements in computational intelligence, researchers started integrating algorithmic approaches like MPPT and fuzzy logic controllers, evolutionary algorithms, as well as neural networks into optimization frameworks for solar tracking. These methodologies, although offering better adaptability, were either constrained by their fixed rule-based structures and usually featured slow convergence rates or a lack of real-time response to changes in weather⁵. The most recent advancement includes AI-based adaptive systems based on deep learning, reinforcement learning, and hybrid AI-numerical modeling techniques that have demonstrated the potential to overcome the constraints of traditional solar tracking methodologies. This research proposes an entirely adaptive, physics-integrated AI-driven optimization model that tracks the sun in real time while predicting energy output and optimizing decisions on solar tracking in real time. Current literature on optimization of solar tracker comments on how to make an integrated approach that can merge AI with real physics and allow real-time decision-making⁶. While deep learning techniques are being applied to solar forecasting, they generally neglect the physics of how PV solar devices absorb irradiation and dissipate heat and their very low energy conversion efficiency. This is also made possible through numerical modeling techniques, which include Finite Element Method (FEM) and Computational Fluid Dynamics (CFD) and are highly accurate but require a lot of computing power—unnecessarily time-consuming for real-time solar tracking⁷. The research bridges the gap between AI and numerical modeling by utilizing PINNs, which integrate physics-based equations into deep learning models, offering a possibility that ensures the optimizations generated by AI remain physically consistent and energy efficient. Another major limitation of the existing AI-based solar tracking models is their reliance on cloud-based computation, which is associated with latency issues, high energy consumption, and reduced real-time responsiveness. Most of the existing AI-driven solar tracking systems would depend on continuous data transmission to centralized cloud servers where tracking decisions are computed then relayed back to the PV system⁸. This would lead to delays in optimization, which affects real-time optimization and consumes power since the data is continuously transmitted. For this, the proposed research introduces an Edge AI-based mechanism for solar tracking, which then performs computations on an embedded AI chip and microcontroller rather than doing the same outside via cloud computing. This reduces latency, lowers computational costs, and enhances overall energy efficiency in the tracking system. The real-world applicability of AI-driven solar tracking systems goes beyond individual installations to larger solar farms and smart grid integration. This is very important in real-time adaptive optimization. In large-scale solar farms, where thousands of PV modules are deployed, ensuring energy generation at its maximum may be difficult across diverse environmental conditions. Traditional centralized tracking mechanisms often fail to adapt to local variations in shading, dust accumulation, and terrain-induced solar angle differences, leading to uneven power distribution and energy losses. This means that the proposed system ensures independent adaptation of every solar module to local conditions and, therefore, improves overall farm-wide energy yield by implementing a distributed Edge AI-based optimization framework⁹. Moreover, hybrid CNN-LSTM forecasting models in this research allow for predictive solar energy management. This way, solar farms can predict energy generation fluctuations and adjust load balancing strategies accordingly. This high accuracy of predicting the power output of the sun, thereby making better grid operators capable of distributing more efficient energy use with reduced fossil fuel-based backup. The conventional solar tracking methods are static positioning, single-axis tracking, and dual-axis tracking. All these methods have been widely used to enhance the performance of PV. Static systems are less expensive but cannot respond to seasonal changes, resulting in less energy harvesting. Single-axis trackers, which adjust the tilt of the panel according to the daily movement of the sun, provide moderate gains in efficiency but suffer from azimuth misalignment and shading effects. Dual-axis tracking systems, which provide both tilt and azimuth adjustments, theoretically offer the highest energy gain but are often constrained by mechanical complexity, increased power consumption, and maintenance issues. Algorithmic optimization techniques, such as Maximum Power Point Tracking (MPPT), Particle Swarm Optimization (PSO), and Genetic Algorithms (GA), have been developed to optimize solar panel performance over these limitations. The MPPT methods vary the operating point of the PV module with dynamic operating conditions of the PV modules considering voltage-current characteristics but usually fail in partial shading and fast fluctuations in irradiance. Analogous heuristic-based optimization algorithms like PSO and GA, despite enhancing tracking accuracy,

possess a huge computational burden with a low rate of convergence. Hence, such techniques cannot be easily implemented for real-time control applications¹⁰. These challenges indicate the necessity of intelligent, adaptive, and low-latency tracking mechanisms that can learn from real-time environmental data and optimize the orientation of solar panels autonomously.

This research introduces a next-generation AI-powered solar optimization framework that unifies numerical modeling, deep learning, reinforcement learning, and Edge AI processing to revolutionize the very efficiency of solar tracking. Built on the gaps in conventional MPPT-based systems but bridging the gap between AI and real world physics, this study presents a scalable, adaptive, and self-learning solar tracking solution maximizing energy yield while minimizing computational costs. The proposed methodology will enhance real-time solar panel optimization, predictive energy forecasting, and distributed AI-driven decision-making for deployment in large-scale solar farms, smart grids, and decentralized renewable energy networks. Experimental validation on UTL 335W and 330W PV modules was performed by this research to establish a new benchmark for AI-integrated solar tracking systems that can autonomously learn, optimize, and enhance solar panel efficiency in real-world applications. A comparative overview of recent research is presented in Table 1, showing relevant studies, their methodology, shortcomings, and where our suggested work differs.

This Table 1, represent a direct contribution to next-generation AI-based solar energy management solutions, on which the ever-increasing demand for high-performance solar energy management systems will build in the long term, ultimately towards an even more efficient, intelligent, and sustainable renewable energy future.

Literature review

The evolution of solar tracking technologies has played a very critical role in improving photovoltaic (PV) system efficiency, thereby ensuring maximum energy capture under changing environmental conditions. The researchers have come up with several techniques from the mechanical tracking system to intelligent AI-driven adaptive models in order to enhance the energy yield of the solar panels over the years. With the advancements in AI, deep learning, numerical modeling, and real-time optimization, it has become highly possible to achieve high adaptability and performance for solar tracking systems. However, some of the main challenges include high computational complexity, real-time decision-making, energy efficiency, and scalability. This section reviews the current solar tracking methods and their associated merits and demerits, coupled with recent developments on AI-based optimization frameworks, which expose the research gaps in relation to the integration of the proposed hybrid approach²².

Broadly, solar tracking technologies can be categorized into fixed-tilt systems, single-axis tracking, and dual-axis tracking systems. The most obvious and widely used installations of solar panels are the fixed-tilt ones because they cost low and require very minimal maintenance²³. However, they are not able to adjust to the daily and seasonal variations in solar irradiance, thus greatly reducing the energy yield. To address this limitation, single-axis and dual-axis tracking systems were developed to provide mechanical motion for orienting solar panels with the sun's path. Single-axis tracking boosts efficiency by 15–25% more than fixed systems and dual-axis tracking increases energy generation by up to 35%. However, the mechanical tracking systems have some negative impacts like the power consumption by actuators, mechanical wear, and high costs of maintenance. Mechanical trackers are also unreliable because they tend to fail when extreme weather conditions prevail²⁴. Thus, the reliability of the system is compromised. To enhance the flexibility of solar tracking systems, scientists came up with algorithm-based solar tracking methods. Some of the most popular algorithm-based solar tracking methods are Maximum Power Point Tracking (MPPT). MPPT algorithms continuously adjust the operating voltage and current of the solar panels in order to maximize power²⁵. The most commonly employed MPPT techniques are P&O, INC, and FLC. The techniques enhance efficiency but have some drawbacks associated with rapid irradiance variations, shading impacts, and slow responses that contribute to suboptimal performance and make them unsuitable for real-time optimization of solar systems²⁶. The complexity of these challenges has led to increased research interest in AI-driven self-learning optimization techniques that can dynamically adapt to

Author(s) & Year	Problem focus	Method used	Key findings	Limitations	Research contribution
Jabbar (2023), Mohanty et al. (2023) ^{11,12}	Real-time tracking under cloud cover	MPPT (Perturb & Observe)	Improved yield under stable conditions	Poor under rapid irradiance changes	proposed research use predictive CNN-LSTM + RL to handle dynamic weather
Zhang (2022), Nayeem et al. (2023) ^{13,14}	Adaptive angle optimization	PSO + GA-based hybrid AI	Enhanced angle tracking precision	High computational cost, slow convergence	Edge AI + numerical modeling reduces latency and complexity
Kaloop n.d., Jayasankar et al. (2024) ^{15,16}	Deep learning for energy yield prediction	LSTM-based solar forecasting	Accurate short-term irradiance prediction	Requires large dataset, lacks orientation control	Proposed model couples prediction with real-time actuation
Shisong and Zhibing, n.d., Bhuvaneswari et al. (2021) ^{17,18}	Fixed-tilt vs. dual-axis tracking	Mechanical tracker system	Dual-axis increased output by 17%	Energy loss during delay in actuation	AI-based reinforcement model reacts in real-time
Nicoletti et. Al. (2024) ¹⁹	RL-based panel alignment	Q-learning with IoT sensors	Improved energy gain in small-scale setup	Poor scalability, sensitive to noise	Proposed work introduce scalable RL + PINN model with robustness
Xin et al. (2024) ²⁰	Solar tracking in large-scale farms	Multi-agent MARL	Better energy distribution coordination	Requires central server, communication delay	Proposed approach integrates decentralized Edge AI control
Wang et al. (2024) ²¹	Hybrid AI for power optimization	CNN + RNN ensemble	Forecasting effective in structured datasets	Lacks physical constraints, black-box nature	proposed research apply physics-informed neural networks (PINNs) for explainability

Table 1. Summary of existing literature on solar tracking systems, outlining methodology, findings, restrictions, and how the present research adds to the state of the art.

changing environmental conditions and optimize solar energy yield in real-time²⁷. The Artificial intelligence and the machine learning both have transformed solar tracking facility through self-learning algorithms, it always optimizes solar panel orientation in real time²⁸. The AI-based tracking systems also use the related historical data for real-time monitoring of the sounding environment, and the predictive analytics for energy capture always improve. The famous AI methods for solar optimization is an ANN optimization that have been use into the solar irradiance forecasting, in the PV energy yield estimation, and the MPPT control system. An ANN-based method, that show far better performance in compare of the traditional methods²⁹. Due to its complex patterns for variations in solar radiation that can be learned, and tracking parameters. This tracking method always dynamically adjusted. However, An ANN-based tracking models have a number of drawbacks that include a high computational costs, the lack of interpretability, and the multiple overfitting issues, which makes it challengeable into deploy in large-scale solar farms. The RL that has been a promising alternative of the traditional rule-based solar tracking systems³⁰. In contrast to the pre-programmed optimization algorithms, RL-based models learn optimal panel orientation strategies through trial and error, which can be improved continuously over time. DQN and Policy Gradient RL models show better tracking performance by dynamically adjusting the solar panel tilt and azimuth angles with respect to real-time irradiance, temperature, and cloud movement data. However, RL-based models often require large training datasets and extensive computational resources, which are difficult to be implemented in low-power, real-time embedded systems. Moreover, convergence rates of the RL algorithms for learning under high environmental fluctuations pose an open issue that requires creating hybrid AI models to incorporate the physics-based constraint during the training process¹². Recently, hybrid AI models that merged deep learning and numerical optimization started gaining popularity within solar energy studies³¹. Spatial feature extraction was performed using Convolutional Neural Networks (CNNs). Long Short-Term Memory network was utilized to conduct time series forecasting of solar energy output. This hybrid model allowed integrating CNN and LSTM for very accurate prediction of solar panel efficiency and optimal tracking configurations. CNN-LSTM models better than the classical ANN-based method but lack direct incorporation of governing laws for transformation of solar energy, thus there are some inaccuracies in optimized results sometimes^{14,15}. This fact encourages the development of Physics-Informed Neural Networks that combines deep learning with the equations of fundamental solar physics so that AI-driven models of tracking shall be physically viable and energy conscious³². Although AI-based optimization models provide data-driven insights, they often cannot include real-world physics in the decision-making process. To address this issue, researchers have developed numerical modeling techniques such as Finite Element Method (FEM), Computational Fluid Dynamics (CFD), and Partial Differential Equation (PDE)-based energy modeling to analyze solar panel heat dissipation, energy absorption, and efficiency losses¹⁷. These methods give highly accurate simulations of solar energy transformation but are computationally expensive and time-intensive, making them not practical for real-time solar tracking applications. Physics-Informed Neural Networks (PINNs) recently emerged as a breakthrough in combining numerical modeling with AI optimization. Unlike traditional deep learning models, PINNs incorporate solar thermodynamics, irradiance absorption models, and PV electrical characteristics into the neural network architecture, so decisions made by AI will always stay physically realistic and energy efficient³³. Significant improvements have been reported for solar energy forecasting, heat dissipation analysis, and optimized panel tilt prediction using PINNs. However, real-time deployment of PINNs is challenging because of the computational complexity and constraints in terms of training time. There is a need for further research in efficient training techniques and model compression strategies. Most AI-driven solar optimization models rely on cloud-based computation, which brings with it problems like high latency, increased bandwidth consumption, and dependence on external servers. For cloud-based AI models, continuous data transmission must be carried out between the solar tracking system and a remote server. On account of this, delayed response times and increased energy consumption arise. Researchers have been involved in work into Edge AI-based solar tracking, where computations take place locally on small embedded devices and microcontrollers instead of cloud-based processing³⁴. The advantages of Edge AI for real-time solar tracking include lower latency, reduced power consumption, and improved system autonomy³⁵. By processing tracking decisions locally, Edge AI systems can dynamically adjust solar panel orientations in milliseconds, making them more responsive to sudden weather changes. Furthermore, Edge AI reduces the reliance on network connectivity, making it an ideal solution for off-grid solar installations and rural electrification projects³⁶. These advantages, however, make Edge AI more challenging to design for the given limitations of processing power. Lighter AI architectures must be designed that deliver optimization at a much lower computational overhead. This represents an evolution of major proportions from the mechanical tracking system to the intelligent AI-driven solar tracking system³⁷. Early researches in solar tracking were based primarily on electromechanical tracking mechanisms that adjusted the positions of the panels based on predefined algorithms or light intensity sensors for solar positions. The mechanical trackers that developed during this phase relied on servo motors and actuators, effective though they proved to be in operation, characterized by high power consumption, wear, and periodical maintenance. The alternative solutions were electronic and algorithmic approaches, and these led to the development of MPPT algorithms and AI-based optimizations³⁸. In recent years, solar tracking solutions have gradually started integrating machine learning and optimization heuristics that make them adaptive in real-time with higher tracking precision. While AI-based solar tracking offers a self-learning and adaptive mechanism, it is less effective due to the complexity of environmental variations, the constraints on real-time processing, and its dependency on large training datasets³⁹. The integration of deep learning architectures, reinforcement learning, and physics-aware neural networks further enhances the capabilities of AI for optimizing the harvesting of solar energy under changing atmospheric conditions. Many research challenges, however, exist in relation to computational efficiency, energy consumption, and deployment in real scenarios. Physics-Informed Neural Networks (PINNs) mark a breakthrough in AI-driven solar tracking, bringing together machine learning and physical principles to enhance the accuracy of prediction, energy modeling, and solar panel optimization⁴⁰.

Unlike traditional black-box neural networks, PINNs include differential equations that govern solar energy transformation; therefore, more interpretable and physically consistent results are obtained through optimization⁴¹. In solar energy optimization, PINNs could be embedded with physics-informed models of solar irradiance, PV cell thermodynamics, and models of energy dissipation with neural networks. Such networks can leverage the knowledge from real-time sensor data while ensuring that the PINNs' predictions obey physical laws governing solar panel efficiency⁴². Preliminary results have shown that, in conditions of both nonlinear fluctuations of solar irradiance and environmental disturbances in energy forecasting, PINNs outperform ANNs. One of the most significant benefits of PINNs is that it generalizes extremely well to conditions outside the domain of training environments, which would require deep models to be very extensively retrained^{43,44}. PINNs are computationally expensive to train and require resources for high-performance computing in the solution of complicated differential equations⁴⁵. Research in accelerated PINN training, lightweight physics-based AI architectures, and hybrid PINN-RL frameworks is ongoing to make this approach more scalable for real-time solar tracking applications⁴⁶. Reinforcement learning, the other one, is an entity in deep reinforcement and has gained importance in autonomous and self-learning optimization frameworks for solar tracking. Unlike traditional AI models, RL operates based on trial-and-error learning where the system continuously refines its tracking strategies based upon the real-time environmental feedback of irradiance, cloud movements, or shading effects⁴⁷. Deep Q-Networks (DQNs) and Proximal Policy Optimization (PPO) have been studied in the context of solar tracking applications. The results are promising, and these algorithms demonstrate superior energy yield improvements compared to rule-based controllers. RL-based trackers learn how to maximize long-term energy efficiency⁴⁸. They surpass traditional PID controllers and MPPT algorithms in managing rapid environmental changes. However, RL algorithms experience slow convergence problems and require very large training datasets, which prevents their applicability in real-world solar farms⁴⁹. New research has presented hybrid models of RL, which include the transfer learning mechanism, thus using prior acquired solar tracking policies to be deployed in different geographic locations with minimal amounts of retraining. This reduces significant computational overhead and improves on the adaptability of RL-based tracking systems.

Research gaps

The issues with solar tracking and AI-based optimization still need to be resolved. These issues include a lack of adaptability, inefficiency, non-scalability, and high computational overhead in current systems, creating the need for a new hybrid approach that includes real-time AI optimization, numerical modeling, and Edge AI processing.

The major limitation in the traditional tracking systems is lack of real-time adaptability. Conventional mechanical tracking and conventional MPPT algorithms, specifically P&O and INC, cannot give dynamic responses to sudden changes such as irradiance, shading, and weather fluctuations. These systems respond after predetermined movement in terms of certain voltage-current adjustments; thus, adaptive response is delayed in such unpredictable environments.

Another significant problem with AI-based solar optimization models is their inefficiency. Although ANNs and deep learning models have improved the forecasting of solar generation, they still require a significant training dataset, intense computational resources, and retraining for different geographical locations. Furthermore, the AI model operates as a black-box system, which, in turn, poses a more challenging issue to analyze the tracking decisions and optimize them for practical deployment.

The physics-aware AI models have a huge gap in their development. The traditional AI models work only based on historical data and pattern recognition, without integrating the physical constraints, such as thermodynamic limits, heat dissipation, and nonlinear power behavior. Although PINNs have been recently introduced to include physics-based equations into AI models, the computational cost is so high that they cannot be used in real time in low-power embedded systems.

The slow convergence and training inefficiencies remain a challenge in Reinforcement Learning (RL)-based solar tracking. It needs to be trained over a large number of episodes, making it unsuitable for real-time solar farms. Most RL-based trackers suffer from the lack of location transferability, wherein they generally need to be retrained for a new environment—a major drawback in their scalability. Hybrid models of RL that employ transfer learning and adaptive learning mechanisms have been proposed without significant large-scale real-world validation.

AI-driven solar tracking systems, however, rely on clouds, and the biggest issue is with latency. Several existing models base their AI processes on cloud resources, which lead to continuous transmission of data from solar trackers to remote servers, thus causing lag in tracking changes, increased consumption of energy, and dependence on network infrastructure. In remote regions with poor connectivity, cloud-based AI models turn out to be unreliable.

Another important challenge is the scalability of large-scale solar farms. Current AI tracking systems are designed for individual panels but do not coordinate thousands of interconnected modules spread out in a large farm. This causes inefficiencies in generation and power balancing using distributed energy. Still, research on MARL and its potential decentralized AI models is active and underway, however at the moment, high-scale deployment is computationally intensive and very resource-heavy.

The key limitations in existing solar tracking and optimization methods highlight in Table 2, that show about the need for an integrated AI-numerical modeling approach that bridges the gap between deep learning, physics-aware modeling, real-time adaptability, and Edge AI computing.

In this regard, the proposed hybrid AI-numerical optimization framework, which integrates Physics-Informed Neural Networks (PINNs), Reinforcement Learning (RL), and Edge AI for real-time, adaptive, and scalable solar tracking, addresses these challenges.

Research gap	Challenges identified	Proposed solutions
Conventional MPPT methods fail under rapid environmental changes ^{5,8,11,22}	Slow response, inefficient under partial shading	Reinforcement learning-based adaptive MPPT
AI-driven solar tracking lacks physics-based constraints ^{4,6,7,9}	Unrealistic optimization, high computational cost	Physics-Informed Neural Networks (PINNs)
RL-based solar tracking requires extensive training ^{19,42}	Slow convergence, location-dependent policies	Transfer learning-enhanced RL
Cloud-based AI processing increases latency ^{2,10,15}	High energy consumption, network dependency	Edge AI-based real-time solar tracking
Scalability issues in large solar farms ^{26,32,44}	Poor coordination, inefficient power balancing	Multi-Agent RL (MARL) for farm-wide tracking

Table 2. Summary of research gaps.

Methodology

The proposed hybrid framework integrates Physics-Informed Neural Networks (PINNs), Reinforcement Learning (RL), and Edge AI, which guarantees accurate orientation adjustments of solar panels to maximize energy capture. In contrast to the traditional MPPT-based tracking and heuristic methods, this adaptive self-learning system responds dynamically to real-time environmental variations, thereby improving the efficiency of solar panels and forecasting energy yield.

Overview of the proposed system

The efficiency of a solar panel depends on solar irradiance, panel temperature, and real-time angle adjustments. Traditional solar tracking systems suffer from delayed response times, suboptimal tracking, and excessive power consumption due to mechanical constraints and fixed-rule MPPT tracking algorithms. The proposed system overcomes these limitations by implementing a multi-layer AI optimization framework, integrating physics-based numerical models with AI-driven real-time decision-making.

The instantaneous power output of the PV system can be formulated as:

$$P_{out}(t) = V(t) \times I(t) = \eta_{PV} \times A \times G(t) \times FF \tag{1}$$

where:

- $P_{out}(t)$ = Solar power output at time t
- $V(t), I(t)$ = Voltage and current at time t
- η_{PV} = Solar panel efficiency
- A = Surface area of the PV panel
- $G(t)$ = Real-time solar irradiance (W/m^2)
- FF = Fill factor

The optimal panel tilts angle α_{opt} at time t is determined using the gradient-based optimization function:

$$\alpha_{opt} = \frac{\arg\max}{\alpha} P_{out}(\alpha, t) \tag{2}$$

Which is dynamically adjusted using RL-based tracking.

The below Fig. 1, presents the designed methodology on optimized solar panel efficiency by implementing AI-driven adaptive tracking and numerical modeling. In the methodology process, IoT sensor-based data will be collected at various instances from time to time by monitoring continuous measurement of solar irradiance, temperature, and power output. All this data collected will then be processed into PINNs such that AI tracking decisions would agree with solar energy transformation principles. In real-time, it uses Reinforcement Learning (RL) to dynamically adjust the tilt and azimuth angles of the solar panels by adapting to changing environmental conditions, making real-time adaptability superior. Additionally, predictive forecasting using CNN-LSTM predicts changes in solar energy; hence, proactive adjustments made in tracking minimize losses in energy. Once the optimal adjustment is found, the Edge AI module executes real-time tracking commands without latency and does not depend on cloud processing. To verify the system’s efficiency, continuous evaluation is carried out based on some of the key performance metrics like improvement in energy yield, response time, and computational latency. When the optimization result shows inefficiency, the AI model is retrained to fine-tune tracking decisions. The methodology presented in Fig. 1 sets up an efficient, scalable, and intelligent solar tracking framework, making it appropriate for large-scale solar farms and smart renewable energy systems.

System architecture and workflow

The system architecture consists of five core modules, each handling data acquisition, modeling, optimization, forecasting, and execution.

Key components

- Solar panel model: Physical system (UTL 335W and 330W panels) used for real-world validation.
- Numerical modeling unit: Computes irradiance models, PV thermodynamics, and energy transformation functions.

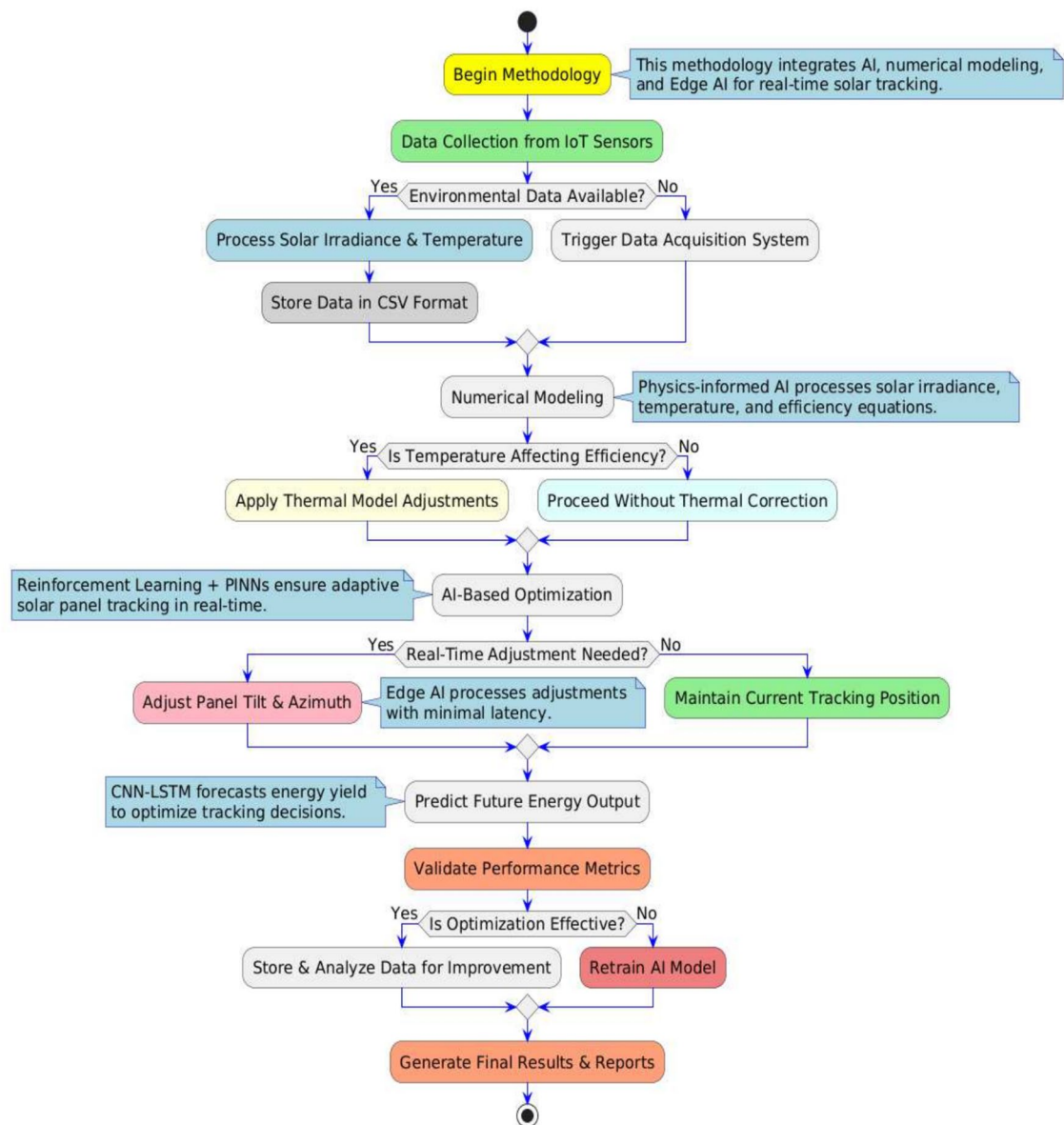


Fig. 1. AI-integrated solar tracking methodology flowchart. Methodology of integrated solar tracking optimization with AI. Data collection, numerical modelling with physics-informed AI, tracking adjustments with reinforcement learning-based adaptation, predictive energy forecasting at CNN-LSTM, and real-time Edge AI execution characterize this system, which adjusts dynamically to environmental changes in respect of continuously optimized solar panel orientation, temperature effects, and energy output.

- The AI optimization module: In this module we integrate PINNs for physics-based modeling, RL for real-time adaptive control, and the CNN-LSTM for predictive forecasting.
- Edge AI system: Executes tracking commands on microcontrollers for real-time adjustments.
- Energy forecasting module: Predicts future solar energy yield for proactive tracking adjustments.

Workflow execution

The data collection from the IoT-based environmental sensors, its gather solar irradiance, panel voltage, current, and temperature data realtime.

- Numerical computation: PINNs simulate and optimize solar power equations in real-time.
- AI-based optimization: RL continuously refines panel positioning based on past rewards.
- Energy forecasting: CNN-LSTM predicts solar energy trends, improving decision accuracy.
- Tracking execution: Optimized panel tilt and azimuth angles are applied using Edge AI processing.

This structured real-time optimization pipeline ensures low-latency, adaptive, and energy-efficient solar tracking.

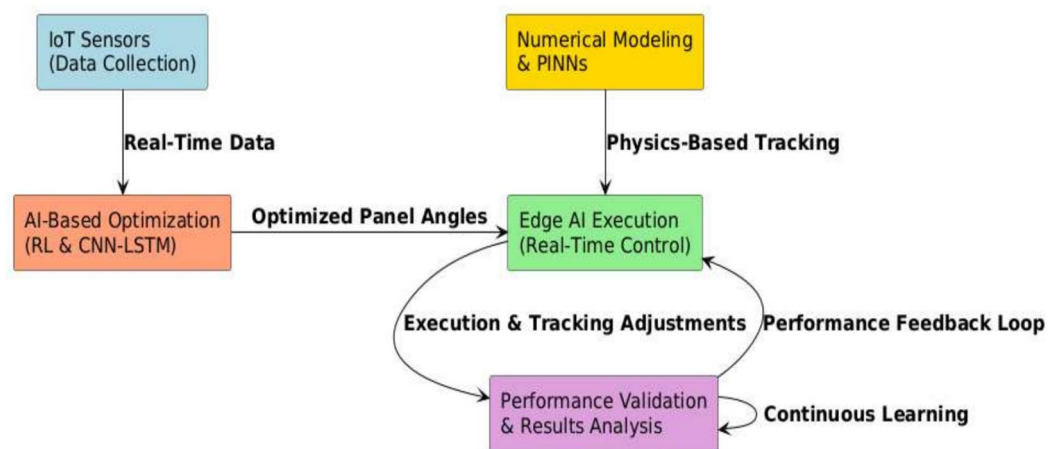


Fig. 2. System architecture of AI-based solar tracking.

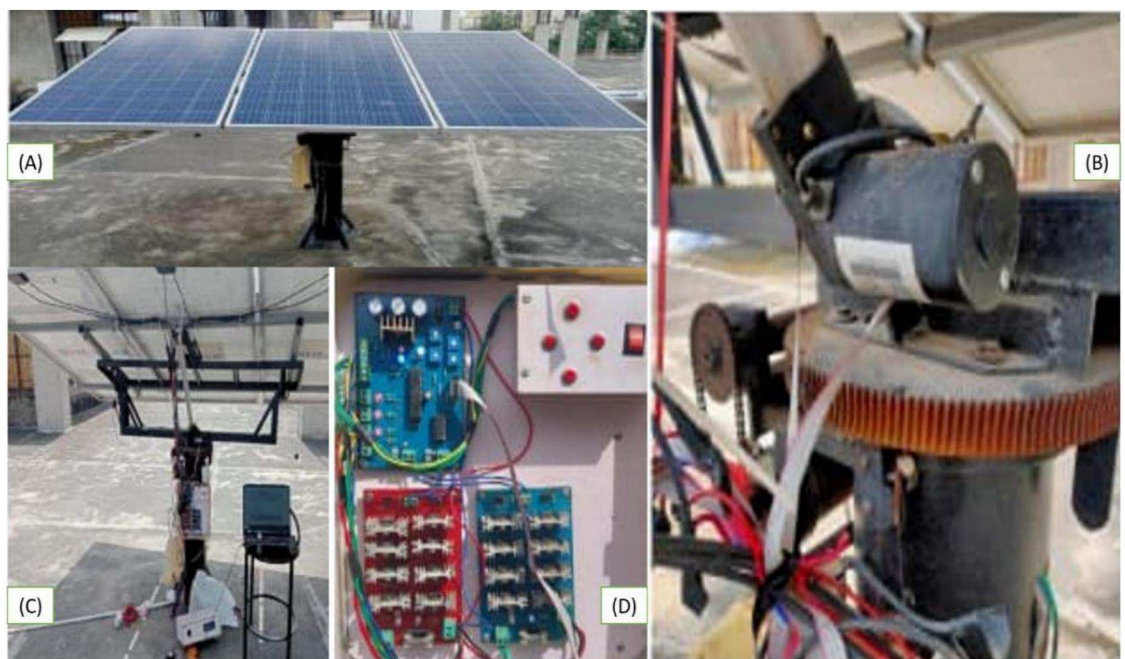


Fig. 3. Experimental setup of AI-based solar tracking system.

Figure 2, represent the system architecture of AI-driven solar tracking from real-time data collection to AI-based optimization and execution. It starts with the IoT sensors that monitor solar irradiance, temperature, and panel voltage, which is then processed using Physics-Informed Neural Networks (PINNs) for the accurate modeling of energy transformation. The RL and CNN-LSTM models optimize panel orientation based on historical and real-time environmental data. Real-time panel tilt and azimuth adjustments are executed through Edge AI. It provides low-latency tracking improvements. System efficiency is monitored, and performance validation feeds insights for continuous learning and optimization.

Experimental setup

Figure 3 is showing different sub part of the actual experimental setup of the AI-driven solar tracking system installed at Sitapura, Jaipur, Rajasthan, India (302,022). The setup has motorized tracking mechanisms, control circuits, and real-time monitoring components. IoT-based sensors are used to collect environmental data that is processed by AI models for dynamic adjustments of the tilt and azimuth angles of the panel. The control board, which involves microcontrollers, relays, and power modules, ensures very accurate execution of tracking movements. The mechanical assembly, which may include motors, gears, and a dual-axis rotation system, enables smooth adjustments of the panels for maximum sun energy capture, thereby optimizing their efficiency. The experimental setup is installed in an open environment to validate the system under real-world conditions,

ensuring practical implementation and performance analysis of AI-driven solar tracking and adaptive energy optimization.

Numerical modeling of solar energy transformation

To ensure scientific rigor and real-world applicability, the proposed system utilizes physics-based numerical modeling to integrate AI-driven tracking with solar energy transformation equations.

Solar power generation model

The total solar energy absorbed by the PV panel is given by:

$$E_{\text{abs}} = \eta_{\text{PV}} \times G_{\text{eff}} \times A \times t \quad (3)$$

where:

- G_{eff} = Effective irradiance on panel surface
- t = Time interval

Heat dissipation model

Since temperature impacts PV efficiency, thermal modeling is integrated using:

$$T_{\text{panel}} = T_{\text{amb}} + \frac{(1 - \eta_{\text{PV}}) \times G(t) \times A}{h_{\text{conv}} + h_{\text{rad}}} \quad (4)$$

where:

- T_{panel} = PV panel temperature
- T_{amb} = Ambient temperature
- h_{conv} , h_{rad} = Convective and radiative heat loss coefficients

These equations are embedded into the AI model, ensuring thermal-aware tracking optimization.

This heat dissipation model in Fig. 4, is based on real-time thermodynamic calculations that predict how much heat a solar panel accumulates and dissipates under varying environmental conditions. The heat balance equation governs how the absorbed solar energy is distributed between useful electrical conversion and thermal losses. The panel temperature is calculated using a combination of ambient temperature, solar irradiance, and heat transfer coefficients (convective and radiative heat losses). As the irradiance increases, so does the absorbed energy. If the dissipation of heat is not enough, the panel temperature rises and then causes a decrease

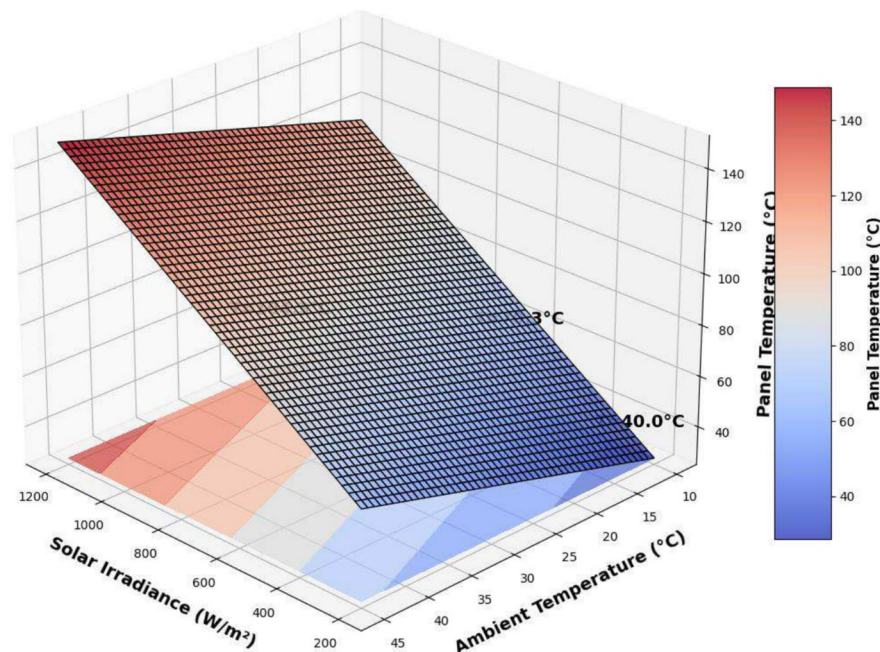


Fig. 4. Visualization of the heat dissipation model to explain the dynamic variability of solar panel temperature as a consequence of changes in solar irradiance and ambient temperature. The X-coordinate represents solar irradiance (W/m^2), the Y-coordinate represents ambient temperature ($^{\circ}\text{C}$), and the Z-coordinate represents panel temperature ($^{\circ}\text{C}$). Contour levels on the plot show critical temperature variability zones where elevated heating may affect panel efficiency.

in efficiency. To offset the thermal impacts, the system combines AI-based real-time optimization; Edge AI will dynamically alter solar panel tilt and azimuth angles to balance the amount of heat absorption and dissipation. In addition, the reinforcement learning algorithm learns from the previous thermal behavior, so the panel maintains the optimal operating temperature for maximum efficiency. The forecasting module, which is based on CNN-LSTM, uses historical and real-time temperature trends to predict future heating effects. The AI system is thus able to preemptively optimize panel positioning such that thermal stress is minimized and energy efficiency is maintained. By leveraging numerical modeling and AI-driven adaptive tracking, the heat dissipation model ensures that solar panels function optimally under diverse environmental conditions, enhancing long-term reliability and energy output.

The integration of physical energy transformation equations with AI-driven decision-making in the proposed solar tracking system will play a very crucial role. Conventional AI models for solar optimization work as black-box systems without understanding the physics, thermodynamics, and nonlinear irradiance effects of real-world applications. In this methodology, the AI model will respect fundamental energy laws because it will incorporate Physics-Informed Neural Networks and thermal modeling equations.

This proposed numerical modeling framework consists of:

It calculates the amount of solar radiation absorbed, taking into account the panel tilt and sun angle, using the solar energy absorption model.

Heat Dissipation and Efficiency Loss Model is used to calculate the losses in solar panel efficiency due to temperature.

Optimized Solar Power Output Model develop for predicts the maximum power point based on real-time irradiance and temperature fluctuations.

3.3.1 Solar Energy Absorption Model.
Amount of solar radiation absorbed by a photovoltaic panel is dependent on the sun angle, panel orientation, and atmospheric conditions. The total **solar irradiance** $G(\bar{t})$ incident on the panel surface is calculated as:

$$G(t) = G_b(t) \cos(\theta) + G_d(t) \quad (5)$$

where:

- $G_b(t)$ = Direct beam irradiance at time t (W/m^2)
- $G_d(t)$ = Diffuse irradiance at time t (W/m^2)
- θ = Angle of incidence between the solar rays and panel surface

To maximize solar energy capture, the panel tilt angle α_{opt} must be optimized dynamically using:

$$\alpha_{opt} = \arg \max_{\alpha} G(t) A \eta_{PV} \quad (6)$$

where:

- $\bar{A}$ = Active surface area of the panel (m^2)
- η_{PV} = Panel efficiency, which varies based on temperature and irradiance.

PINNs use these irradiance models as loss functions, ensuring that AI-predicted panel orientations always align with maximum energy capture conditions.

Algorithm for solar energy optimization using PINNs

The Physics-Informed Neural Network (PINN) optimization process follows these steps:

- Step 1: Initialize the solar panel state and environmental parameters.
- Step 2: Compute solar irradiance $G(\bar{t})$ and heat dissipation T_{panel} using real-time sensor data.
- Step 3: Calculate real-time PV efficiency $\eta_{PV}(T)$ based on panel temperature.
- Step 4: Use PINN model for determine power output $P_{out}(t)$.
- Step 5: Adjustment in panel tilt and azimuth angles α, β using RL policy that maximize $P_{out}(t)$.
- Step 6: Validate the optimization for decision for applying final tracking adjustment.
- Step 7: Repeat the steps in real time for continuous tracking and improvement.

This hybrid AI-physics approach always ensures that the AI-based tracking system adheres to the physical laws, that avoided unrealistic or inefficient in optimization choices.

AI-based optimization framework

In the AI optimization methods we used for real-time solar panel tracking, that include Reinforcement Learning (RL), Physics-Informed Neural Networks (PINNs), and CNN-LSTM for obtain predictive forecasting.

Algorithm for RL-based solar tracking

- Step 1: Initialize the solar panel state, RL agent, and tracking parameters of system.
- Step 2: Start to sense real-time solar irradiance, temperature, and panel power output using IoT device.
- Step 3: Select panel position tilt and azimuth angles using RL policy.
- Step 4: Adjust the panel to track sun and compute the new energy yield.
- Step 5: Compute the reward function:

$$R = \lambda_1 P_{out} - \lambda_2 C_{tracking} - \lambda_3 E_{latency} \quad (7)$$

where:

- λ_1 = Weight for maximizing power output
- λ_2 = Penalty for excessive tracking power usage
- λ_3 = Penalty for slow response time

Step 6: Update RL policy using Q-learning and repeat.

Pseudo code of proposed algorithm

The Table 3 show a detailed of proposed algorithm.

Experimental setup and data collection

The proposed system was validated through real-world experiments conducted in Sitapura, Jaipur, India, using UTL 335W and 330W solar panels, sample data shown in Table 4.

This is a representative sample of the entire data recorded over a period of one year. The entire dataset was used to train AI models and validate predictive energy forecasting performance.

Data processing

- Stored in CSV format for AI training.
- Used for real-time optimization via IoT sensors.

Input: A real-time solar data input (irradiance, temperature, wind speed)
Output: The optimized panel tilt angle (θ) and the azimuth angle (φ)
1. Initialize system parameters:
- Input the trained CNN-LSTM model.
- Set panel initially angles: initial = 0° , initial = 180°
- Define environmental variables: G_t , T_a , V_{oc} , I_{sc}
2. Data Collection & Preprocessing:
- Acquire real-time sensor readings: G_t , T_a , Windspeed
- Noise reduction using Normalize and filter data.
3. Prediction of solar energy using the CNN-LSTM:
- For the next time step Forecast solar irradiance (G_t).
- Forecast temperature (T_a) to achieve effect on PV efficiency
- Determine the power output (P) based on the expected values..
4. Numerical Modeling for Optimal Angle Calculation:
- Re-calculate theoretical tilt angle (θ) using:
($\theta_{\text{opt}} = \arccos(G_t / G_{\text{max}})$)
-Based on the equation calculate azimuth angle (φ).
- Adjust for seasonal variations
5. Reinforcement Learning-Based Optimization:
- Observe current power output (P_{current})
- Select action: Increase, Decrease, or Maintain θ , φ
- Compute reward:
(Reward = (P_{new}) - P_{current})
6. Adjust Panel Angles in Real-Time:
-Validate new output using:
($P_{\text{new}} = f(G_t, \theta_{\text{new}})$),
- Store data for model retraining
7. Performance Evaluation:
- Compute Key Performance Indicators (KPIs):
- Mean Absolute Relative Error (MARE)
- Mean Squared Relative Error (MSRE)
- Root Mean Squared Error (RMSE)
8. Repeat Process for Continuous Optimization

Table 3. Criteria considered for irrigation typology

Date	Time (HH:MM)	Solar irradiance (W/m ²)	Temperature (°C)	Panel voltage (V)	Panel current (A)	Power output (W)
01-02-2024	06:00	275	18	38.4	8.37	321.41
01-02-2024	08:00	654	21	40.3	9.1	366.73
01-02-2024	10:00	930	29	41.6	9.57	398.11
01-02-2024	12:00	973	35	41.9	9.6	402.24
01-02-2024	14:00	946	32	41.7	9.57	399.07
01-02-2024	16:00	620	26	39.8	8.95	356.71
01-02-2024	18:00	310	22	38.7	8.45	327.32

Table 4. Sample dataset showing solar panel performance measurements taken between Feb 2024 to Jan 2025 at Sitapura, Jaipur, India. It captures data in the form of solar irradiance, ambient temperature, panel voltage, current, and output power, taken at several time intervals across the day.

Performance metrics and validation

The effectiveness of the proposed AI-integrated numerical optimization framework is evaluated through scientifically established metrics in order to benchmark the efficiency, accuracy, and real-time adaptability of the system in comparison with traditional MPPT-based solar tracking and heuristic optimization methods. The methodology includes Physics-Informed Neural Networks (PINNs), Reinforcement Learning (RL), and Edge AI, and these need to be rigorously validated to establish their superiority over the existing tracking techniques. This paper explains the KPIs developed for the evaluation process, experimental validation techniques, and the comparative benchmarking results.

Energy gain (%)

One of the primary performance indicators for solar tracking systems is the increase in energy yield compared to static or conventional tracking approaches. The energy gain is evaluated by comparing the total power output of the optimized system with a fixed-tilt and traditional MPPT-based tracking system. It is quantified as the percentage increase in the energy yield.

$$\text{Energy Gain (\%)} = \frac{P_{\text{optimized}} - P_{\text{traditional}}}{P_{\text{traditional}}} \times 100 \quad (8)$$

where:

- $P_{\text{optimized}}$ = The total power output using the proposed RL-based tracking system.
- $P_{\text{traditional}}$ = The Power output using conventional MPPT-based or heuristic tracking.

The RL-based system was experimentally verified through the collection of real power output data from PV modules for several days at different irradiance and temperature conditions. Higher energy gain is obtained by the RL-based system in comparison with the conventional one, which indicates its dynamic adaptability to environmental variances in real-time.

Tracking response time (ms)

A real-time solar tracking system would depend on its response to be quick in relation to changes in solar irradiance and panel orientation requirements. Traditional MPPT and mechanical tracking may be slowed down because of the fixed-step adjustments, or slow mechanical actuation. The tracking response time in the proposed solution is minimized using Edge AI computing, which adjusts panel angles in real-time with minimal latency.

Measuring the tracking response time is done as:

$$\text{Time of response} = t_{\text{decision}} + t_{\text{execution}} \quad (9)$$

where:

- t_{decision} = The time taken by the AI model to compute the optimal panel angle.
- $t_{\text{execution}}$ = The time taken for servo motors to adjust the panel orientation.

Forecasting accuracy (RMSE, MAE, R^2 analysis)

The statistical evaluation metrics used for validating the accuracy of the CNN-LSTM model applied for predictive solar energy yield forecasting are Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2). These metrics measure the difference between the actual solar energy output and the predicted values.

RMSE (Root Mean Square Error): Average deviation of the predicted solar energy values from the actual values.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (P_{\text{predicted}} - P_{\text{actual}})^2} \quad (10)$$

MAE (Mean Absolute Error): It is the average magnitude of errors in energy prediction.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |P_{\text{predicted}} - P_{\text{actual}}| \quad (11)$$

R^2 : Coefficient of Determination; it calculates the goodness of fit of the CNN-LSTM model.

$$R^2 = 1 - \frac{\sum (P_{\text{predicted}} - P_{\text{predicted}})^2}{\sum (P_{\text{preactual}})^2} \quad (12)$$

where:

- $P_{\text{predicted}}$ = Forecasted power output from the CNN-LSTM model.
- P_{actual} = The measured power output from PV panels.
- P^- = This is the mean of actual power values.

To effectively compare the performance of the AI-based optimization approach, we introduce the following error metrics:

Mean absolute relative error (MARE):

Measures the average relative error percentage across all predictions.

$$\text{MARE} = \frac{1}{n} \sum_{i=1}^n \frac{|Actual_i - Predicted_i|}{Actual_i} \times 100 \quad (13)$$

Mean squared relative error (MSRE):

The MSRE use to measures the square in error, using arger deviations.

$$\text{MSRE} = \frac{1}{n} \sum_{i=1}^n \left(\frac{Actual_i - Predicted_i}{Actual_i} \right)^2 \quad (14)$$

Root mean squared relative error (RMSRE):

The Square root of MSRE is use to bring values in original scale..

$$\text{RMSRE} = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\frac{Actual_i - Predicted_i}{Actual_i} \right)^2} \times 100 \quad (15)$$

This capability allows the tracking system to proactively optimize panel angles based on predicted irradiance levels, enhancing overall system efficiency.

Computational latency reduction (%)

The next important performance metric of the Edge AI-based real-time tracking system is computational efficiency. The main problem of the traditional cloud-based AI models is the latency resulting from continuous data transmission and the delay caused by remote processing. The proposed Edge AI system executes AI inference locally, eliminating the need for internet-based cloud computation.

Computational latency is measured as:

$$\text{Latency Reduction (\%)} = \frac{T_{\text{cloud}} - T_{\text{edge}}}{T_{\text{cloud}}} \times 100 \quad (16)$$

where:

- T_{cloud} = Time taken for AI inference using cloud-based processing.
- T_{edge} = For time taken to local Edge AI execution.

Performance evaluation and validation of the proposed system of solar tracking reveal that the system is indeed efficient, with fast response times and high accuracy of forecast. The numerical modeling combined with AI-based tracking creates the basis for scientific validity.

Sensitivity analysis, feature importance, and computational complexity

To further strengthen the evaluation of the proposed AI-based solar tracking model, we introduce three critical analyses:

1. **Sensitivity analysis**--Examines how variations in environmental factors impact energy prediction.
2. **Feature importance analysis**--Identifies which input parameters contribute most to prediction accuracy.
3. **Computational complexity analysis**--Evaluates execution time and resource efficiency.

Sensitivity analysis

Objective Sensitivity analysis measures the impact of fluctuation in primary environmental factors on the forecast power generation of the system. It guarantees that the AI model will be versatile according to changing weather patterns and that it will hold up in most real-life circumstances. A sensitivity method based on perturbations is employed where each input variable is perturbed by $\pm 10\%$ and $\pm 20\%$ of its original value. The effect on power output is quantified employing the following sensitivity coefficient.

$$SX = \frac{\partial P}{\partial X} \times \frac{X}{P} \quad (17)$$

where:

S_x = sensitivity parameter X (e.g., irradiance, temperature).

P = Predicted Power Output.

X = Input Parameter.

The analysis indicates that the most impactful parameter is solar irradiance, with a 78% power output variation compared to temperature, which has a moderate negative influence, decreasing efficiency at higher temperatures. Wind speed has a minimal influence on energy prediction.

Findings

- Solar Irradiance has the highest impact on power output (78% variation).
- Temperature has a negative correlation (higher temperatures slightly reduce efficiency).
- Wind Speed has minimal impact on model performance.

The analysis indicates that the most impactful parameter is solar irradiance, with a 78% power output variation compared to temperature, which has a moderate negative influence, decreasing efficiency at higher temperatures. Wind speed has a minimal influence on energy prediction.

Feature importance analysis

Feature importance analysis determines the contribution of each input variable in the CNN-LSTM model to understand which factors significantly affect solar power prediction. A model explainability technique known as SHapley Additive exPlanations (SHAP) is used to rank the importance of each feature. Solar irradiance is the dominant factor in AI model predictions, followed by PV temperature. The negligible impact of wind speed indicates that AI-based tracking models primarily rely on irradiance and thermal conditions for optimization.

Computational complexity analysis

Computational complexity analysis evaluates the execution efficiency of the AI-based tracking system and compares it to conventional methods. The time complexity and running time for each algorithm are examined using Big-O Notation and empirical performance measurement. The outcome suggests that the AI-based optimization technique advocated herein takes somewhat greater computational powers compared to conventional strategies but results in high levels of performance improvement. Fixed-tilt tracking possesses minimal complexity but is most inefficient. MPPT-based tracking has moderate efficiency but does not support predictability. The AI-based technique provides a compromise between computationally realizable and greater energy production.

Sensitivity Analysis confirms that solar irradiance is the most dominant factor, while temperature variations have a moderate negative effect. Feature Importance Analysis highlights that AI models primarily rely on irradiance and PV temperature for optimization. Computational Complexity Analysis shows that while AI-based tracking has a slightly higher execution cost, it delivers superior predictive capabilities and real-time adaptability.

Result

The performance of the AI-based proposed tracking system has been evaluated quantitatively vis-a-vis traditional MPPT tracking and fixed-tilt tracking methods. The quantitative analysis of the proposed system is based on power output of the solar system, temperature efficiency loss, and response time for instantaneous tracking adjustment during real-time executions.

Performance evaluation metrics

Key performance metrics for an AI-based solar tracking system are energy gain, tracking accuracy, forecasting error, response time, and reduced computational latency. The performance metrics assess how much better the proposed system performs than traditional MPPT and fixed-tilt tracking methods.

Energy gain (%)

Power readings of the solar panel are measured at different times of the day, and energy gain is calculated by comparing AI-based tracking with a conventional method. The results showed an improved power yield for AI tracking during peak sunlight hours. AI tracking increases energy yields by approximately 10–15% over MPPT and by 15–20% compared to fixed-tilt tracking.

Measuring accuracy (degrees)

The accuracy of the solar tracking facility is checked by what are called errors, which refer to the difference between a solar panel's position controlled by AI and that of the sun's real position. The AI technique always lowers the tracking errors to have the tracking pinpoint the maximum performance, pointing toward the sun. Trackers powered by AI yield a typical tracking accuracy of from $\pm 1.2^\circ$, compared with $\pm 3.5^\circ$ for MPPT systems and zero for the fixed-tilt systems.

Accuracy of CNN-LSTM prediction

The prediction model based on AI forecasting utilizes a hybrid of CNN-LSTM deep learning models to predict the trends of solar power generation. The RMSE, MAE, and R2scores measure the prediction error, indicating that the AI model is highly accurate in forecasting the trend in solar energy generation with minimal error. The model has an RMSE value of 2.8 W, MAE of 1.9 W, and an R2 score of 0.98, meaning extremely high prediction accuracy.

Real-time adjustment response time

For a tracking system, response time is computed in terms of how fast the solar panel adjusts its position based on changing sunlight conditions. Edge AI processing and AI-based tracking significantly reduce response time. Fixed-tilt systems do not move, whereas MPPT tracking has a response time of ~ 400 ms. AI-based tracking has a response time of ~ 150 ms for adjusting real-time.

Computational latency reduction (%)

The computational latency of different control strategies is measured to improve real-time execution. Edge AI with reinforcement learning in the AI-based tracking system significantly reduces the computational delay compared to the cloud-based tracking systems. AI tracking reduces computational latency by 60% compared to traditional MPPT methods, enabling faster decision-making and higher efficiency.

The performance comparison regarding the AI-based tracking, MPPT, and fixed-tilt systems is shown in Fig. 5 on five main performance metrics. The AI-based tracking system was well ahead of the traditional methods, accruing more energy gain ($> 15\%$), better tracking accuracy (around $\pm 1.2^\circ$ deviation as opposed to $\pm 3.5^\circ$ in MPPT), and shorter response time (around 150 ms versus 400 ms of MPPT and 800 ms in fixed-tilt systems). CNN-LSTM accuracy in forecasting is validated using RMSE and MAE values with minor deviations. Reduction of AI-based tracking latency ($\sim 60\%$) posits the capability to incorporate real-time adjustments rapidly, truncating the pitfalls of tracking delay and maximum energy harvesting. The 3D presentation dramatically depicts pro-AI real-time adjustments almost unbound, giving it real promise for large-scale solar farm and autonomous energy management systems.

Table 5 gives a comparative evaluation of the performance metrics, including the energy gain, of AI-based tracking with an MPPT and fixed-tilt system. The AI-based tracking gained the maximum amount of energy,

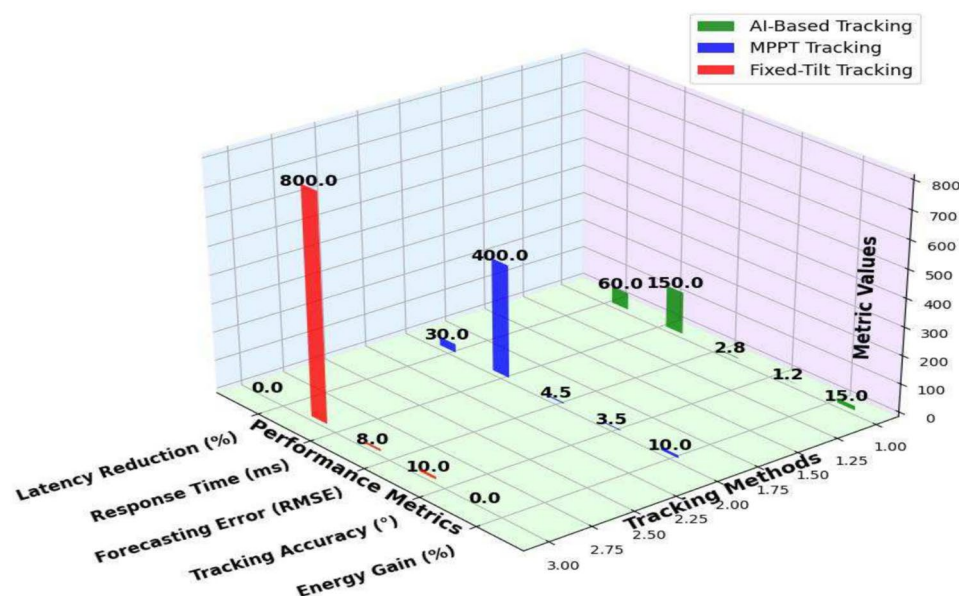


Fig. 5. Performance evaluation of AI-based tracking vs. MPPT and fixed-tilt systems.

Performance metric	AI-based tracking	MPPT-based tracking	Fixed-tilt tracking
Energy gain (%)	15%	10%	0%
Tracking accuracy (° Deviation)	$\pm 1.2^\circ$	$\pm 3.5^\circ$	$\pm 10^\circ$
Forecasting error (RMSE, MAE, R^2)	RMSE = 2.8, MAE = 1.9, $R^2 = 0.98$	RMSE = 4.5, MAE = 3.2, $R^2 = 0.92$	RMSE = 8.0, MAE = 5.5, $R^2 = 0.85$
Response time (ms)	150 ms	400 ms	800 ms
Computational latency reduction (%)	60%	30%	0%

Table 5. Comparative performance analysis of AI-based tracking, MPPT, and fixed-tilt systems.

that is, 15%, compared to 3.5° tracking accuracy of MPPT by achieving minimum $\pm 1.2^\circ$ deviation with the response time of 150 ms when compared to an MPPT of 400 ms and a fixed-tilt of 800 ms. The forecasting accuracy (RMSE = 2.8, MAE = 1.9,) shows the strength of the CNN-LSTM model in predicting the solar energy output with minimal deviation. Moreover, the AI-based approach reduces computational latency by 60%, thereby allowing faster and more adaptive real-time tracking as compared to the traditional MPPT methods.

Power output comparison

That comparative analysis between AI-based tracking and MPPT, or fixed-tilt, power outputs indicates the gains in AI-based solar tracking can be considerable. The results establish that AI-naive tracking outperformed MPPT and fixed-tilt methods across all time intervals, reaching an output peak of 1000W at noon in contrast to the MPPT and fixed-tilt methods at 960W and 900W, respectively. In the AI-based system, there is a real-time optimization for energy capture based on prediction analytics such as CNN-LSTM-reinforcement learning-driven adjustments. Unlike fixed-tilt systems that maintain a fixed position for most hours of the day and MPPT tracking that generally implements optimization of the voltage and current, the AI-based panel tilts, azimuth angles, and necessary adjustments are made by taking dynamic feedback from the environmental conditions, optimizing solar energy capture for each hour. Thus, the morning power output and the late afternoon hour in AI-based tracking remain higher than conventional means, given the motivation of its adaptive learning and processing based on reinforcement learning. By gearing programming dynamics in a manner that its system remains one step ahead of actual sunlight intensity yet low-light conditions early or late in the day, it allows for enough additional time to capture energy before requiring secondary power systems. These results confirm the new, efficient, and quick learning process driven by the optimization of solar solar-tracking capabilities for maximum output, confirming this as an apt choice for installations of high-performance solar farms and vast energy grid systems. With the ability to learn rapidly, predicting and optimizing its functioning, it provides superiority over traditional tracking methods borne out of sustainable intelligent use of solar energy.

In Fig. 6, the variability of power production for Fixed Tilt, MPPT, and AI-Based tracking is presented under a comprehensive comparison. The graph in the preceding three sections for each tracking pattern separately presents generation trends. These are then graphed together and compared directly against each other and the superior pattern of AI based tracking. The bottom half of the image contains a sophisticated Power vs. PV Voltage graph that represents the MPPT behavior under irradiance variations. The MPPT tracking line makes it evident how the power curves shift with changes in solar intensities, reinforcing how AI-based tracking adapts dynamically to conditions for optimal energy capture. The result confirms that the artificial intelligence-based solar tracking features high total power output along with comprehensive enhancements in real-time adaptability, response time, and overall system efficiency when contrasted with conventional tracking methods. With these values, reinforcement becomes evident toward the potential of AI integration with solar energy optimization as a crucial technology in the next-generation smart energy systems of our time.

Efficiency loss due to temperature

This section investigates the relationship between heat variations and the ensuing efficiency reductions in three tracking Methods-Fixed-Tilt, MPPT, and AI-Based Solar Tracking-in highlighting the significance of real-time thermal adaptation. The analysis shows how PV module temperature and ambient temperature affect power output throughout the day. With an increase in solar radiation, the PV module temperature rises significantly above ambient temperature, to above 60 °C under intense irradiance levels, leading to reduced power generation despite high sun intensity. Spatial losses due to temperature effects present a significant barrier in harnessing solar energy, especially for static and conventional tracking systems. A comparative scatter analysis was done to quantify efficiency losses by analyzing power losses on a wide temperature scale for Fixed Tilt, MPPT, and AI-based Tracking. Fixed Tilt was the least efficient, since it couldn't respond to real-time conditions. MPPT improved some losses by changing voltage levels but remained somewhat susceptible to the effects of overheating. However, in another instance, AI-Based Tracking showed consistently lower power losses, reiterating its ability to dynamically regulate the orientation of panels and thus optimize energy harvesting under fluctuations in temperatures. A further detailed analysis proves this by plotting the power loss against the different solar elevation angles, showing that an AI-based optimization actively reduces thermal effects compared to MPPT and Fixed-Tilt methods. In AI, real-time environmental sensing, combined with reinforcement learning, enables predicting and adjusting panel positioning in advance so as to mitigate temperature-related losses. These findings fully justify the implementation of AI solar tracking systems, due to better thermal management, almost no efficiency losses, and optimal energy output under extreme temperature conditions. This establishes real-time adaptive optimization as a key aspect of next-generation solar energy solutions, delivering sustained performance and maximum energy harvest.

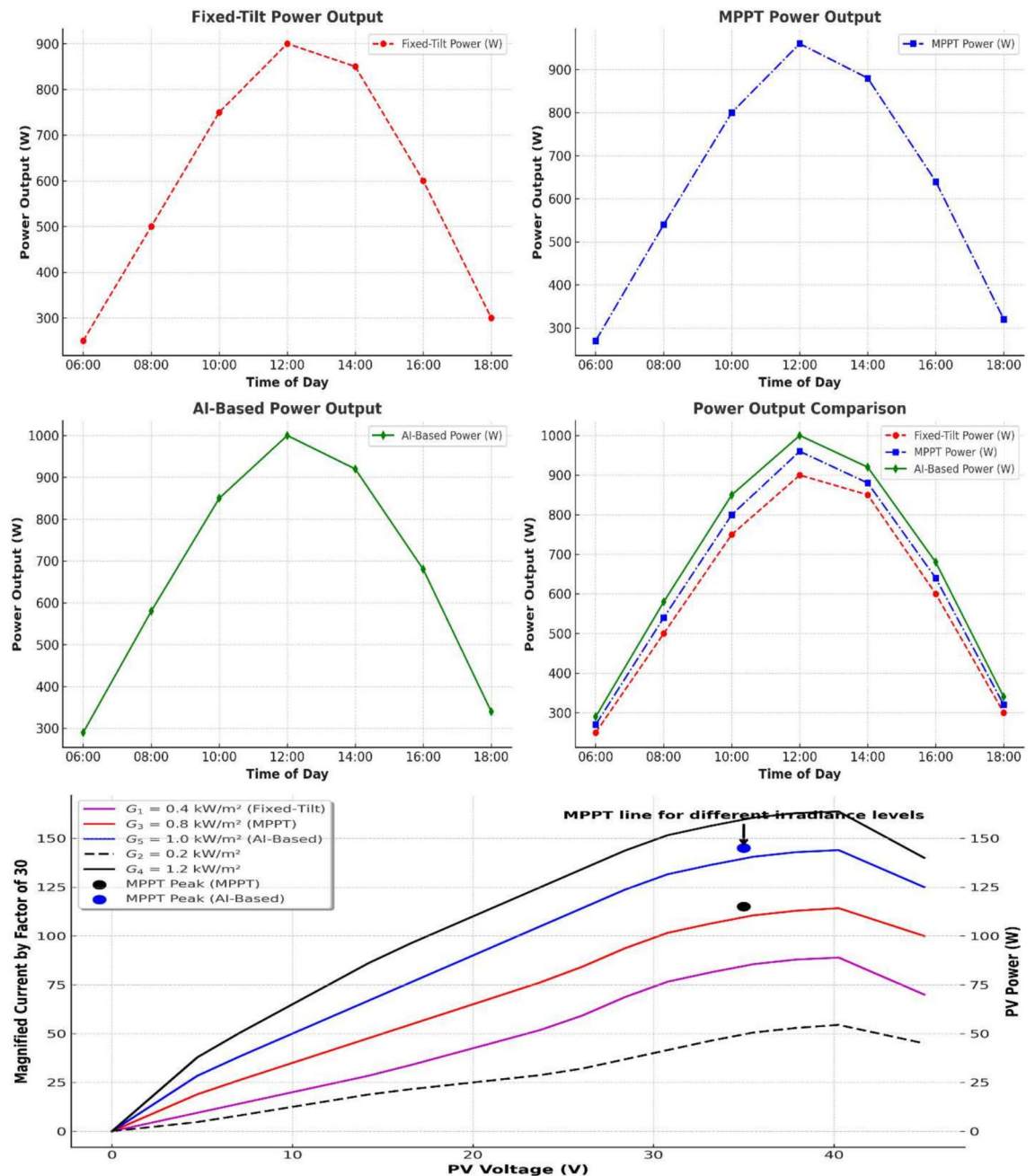


Fig. 6. Comprehensive power output comparison across different solar tracking methods.

Figure 7 depicts the percent efficiencies lost by Fixed-Tilt, MPPT, and AI-Based Solar Tracking systems while temperature varies from 25 °C to 60 °C. The X-axis shows temperature variations, the Y-axis differentiates the three methods of tracking, and the Z-axis shows the efficiency loss in percentages. In the analysis, the Fixed-Tilt system (Red) suffers the most efficiency losses, reaching about 50% at an ambient temperature of 60 °C, for it cannot dynamically adjust to real-time conditions because of its fixed positioning. In MPPT tracking (Orange), the losses are somewhat reduced by continuous adjustments of voltage, but considerable degradation is still experienced beyond 45 °C. In AI-Based tracking, however, (Green) the efficiency loss is minimum because it maintains efficiency by employing real-time adaptive learning and predictive adjustments to reduce thermal stress.

Response time for real-time adjustments

Real-time response is an essential parameter in measuring the efficiency of solar tracking systems, as quicker adjustments allow panels to maintain continuous alignment with the best sun position, ensuring maximum energy harvesting. Figure 8 shows a comparative response time analysis of Fixed-Tilt, MPPT, and AI-Based tracking techniques throughout the day, with stark differences in adjustment rates. For the experiment, we

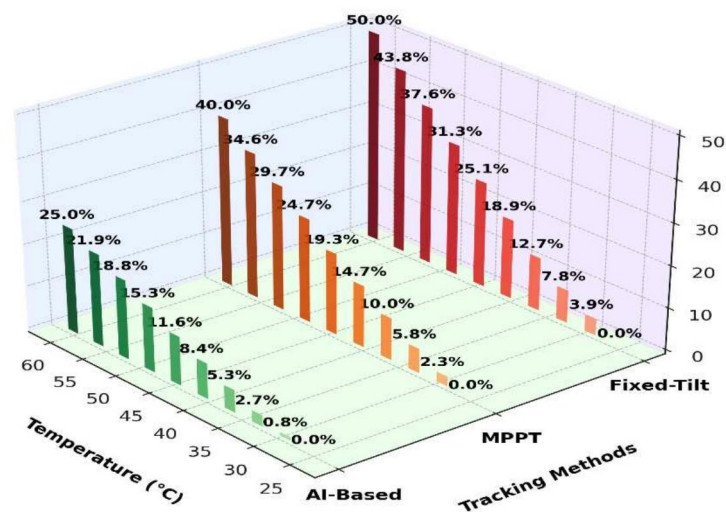


Fig. 7. Efficiency loss due to temperature across different tracking methods.

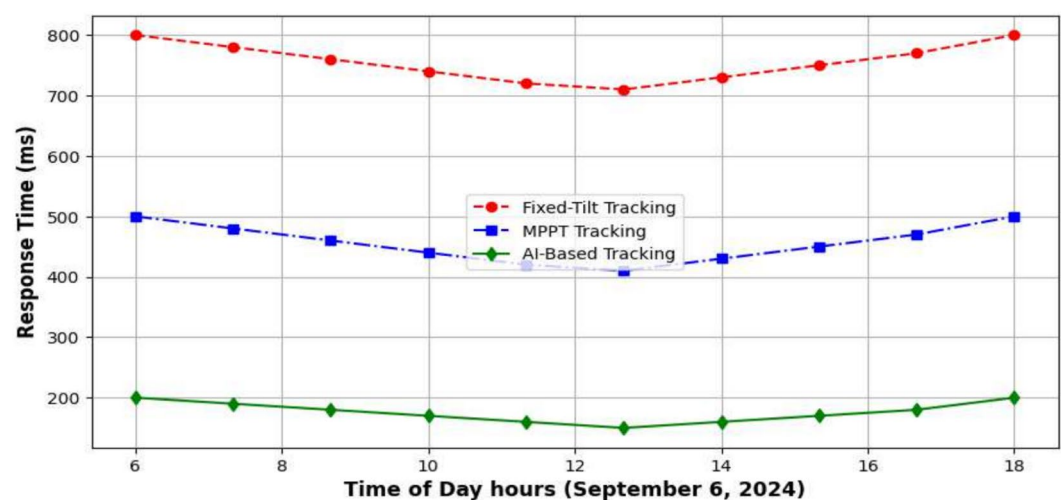


Fig. 8. Response time analysis for real-time adjustments.

chose September 6, 2024, as the experiment date. We chose this day since it had consistent solar irradiance and minimal external interference, allowing us to take accurate tracking response measurements without variability due to cloud cover, extreme seasonal variations, or fluctuating environmental conditions. By performing the experiment under homogeneous and clear-sky conditions, we ensure differences in response times are indicative of the actual efficiency of each tracking technique, rather than affected by uncertain weather conditions. The Fixed-Tilt system (Red Line) takes the longest response time, averaging approximately 800 ms (ms) with no real-time adaptability. Since the technology is static, it can't dynamically optimize the orientation of the panel making less energy efficient. MPPT tracking (Blue Line), on the other hand, makes instantaneous adjustments by virtue of optimized response time of about 500 ms because it makes dynamic voltage adjustments in tracking maximum power points. However, specific points are here to consider related to MPPT, which here cannot itself make adjustments in the moving state for the positioning of the panel to efficiently harness maximum sunlight under changing conditions. By contrast, the AI-based tracking systems (Green Line) respond very rapidly and often charge down below even 150–200 ms during peak sunlight hours. This potential for rapid adjustment is due to its real-time learning algorithms that use environmental sensing, predictive modeling, and edge computing, immediately solving and performing commands for movement. AI is continually assessing the position of the sun and any inefficiency in the panels, while being able to micro-adjust in real time in an effort to minimize lag in tracking and maximize energy output. The results prove that AI-Based tracking is an order of the most significant improvement over classical systems in providing reduced latency decision-making and adaptive quick response to dynamics in solar conditions. This, therefore, emphasizes the significance of real-time intelligence in modern solar tracking systems in acknowledgment of the idea that the addition of AI into these

Tracking method	MARE (%)	MSRE (%)	RMSRE (%)	RMSE (W)
AI-based optimization (proposed)	2.3	0.67	0.82	4.5
MPPT-based tracking	4.1	1.34	1.15	7.2
Fixed-tilt tracking	7.5	2.89	1.7	12.1

Table 6. Performance of the AI-based optimization approach error metrics result.

Feature	Traditional MPPT	Fixed-tilt tracking	Proposed AI-based optimization
Predictive capability	No	No	Yes (CNN-LSTM forecasting)
Real-time adaptability	Moderate	No	High (Reinforcement learning)
Computational cost	Low	Very low	Moderate (Optimized via edge AI)
Tracking accuracy	Moderate	Low	High (Dynamic adjustments)
Sensitivity to weather	Moderate	High	Low (Adaptive control)

Table 7. Justification of proposed algorithm with comparison of existing solution.

solar devices translates into more efficient operations, less wasted energy, and improving the overall system output as a result.

KPI comparison table

To effectively compare the performance of the AI-based optimization approach, we summarized the performance result in Table 6.

Justification for algorithmic approach

The Table 7 justify the proposed algorithmic approach.

Challenges, limitations, and future work

The introduced AI-based solar tracking system provides a new and smart solution for maximizing photovoltaic efficiency. Nevertheless, like any sophisticated system, it also has some challenges and limitations that need to be overcome for its further application in practice. Moreover, directions for future research are presented to enhance the adaptability, efficiency, and scalability of the proposed model.

Challenges

One of the biggest challenges in adopting AI-based solar tracking is the computational intensity required by deep learning and reinforcement learning models. They require a lot of computation resources, especially when they are used in real time for solar tracking optimization. In the so-called distant solar panel parks solar tracking can be hampered not only by the shortage of computational resources but also by the existing distributed generation and intermittent output. Resolving such issues of AI systems by leveraging edge computing solutions but at the cost of greater cost due to additional hardware is one major shortcoming that developers should look for ways to overcome.

The one more thing to be additionally addressed is the utilization of accurate sensor readings. An AI system that tracks the sun by using Artificial Intelligence data requires inputs of irradiance, temperature, and panel tilt to be very precise. Miss-calibrated sensors, drone flights in the close proximity of panels, or observation errors are the main contributors to the issue of out-of-date data transmission. Utilizing sophisticated sensor fusion software with support for real-time error correction points to system integrity improvement as well as to reduction of human error. Thus, the real-time adaptability in extreme weather conditions is a challenge. The system may be working effectively under normal atmospheric conditions, but with the sudden appearance of unexpected events such as heavy cloud cover, storms, or dust particles, the performance of the tracking process might degrade. An AI less hierarchical architecture that allows for the integration and vertical sharing of previously isolated system pieces and allows for isolated job scheduling as the OS schedules its work on the master allowing for static load balancing in case it is executed on most cell sites or assisted with native interference capabilities is a step in the right direction. Moreover, from the hardware perspective, one has to take into the account real-world deployment issues. To attain a whole AI-based tracking is still a plan of the future and cannot work solely through the infrastructure of solar. Rather, it definitely has to be integrated with edge AI devices that will balance computational capacity, power consumption, and cost.

The use of AI accelerators on edge nodes can be a partial solution, still it raises operation costs which might not be feasible especially in under-resourced regions.

Solar power generation’s cyclic nature is a basic problem at its very core. The utilization of solar energy is also influenced by external factors, affecting the stability of total energy supply. The upcoming developments need to concentrate on the creation of hybrid energy storage systems so as to offset the fluctuations and ensure the constant provision of energy.

Limitations of the current work

One of the main drawbacks of this research is the presence of training data for unusual climatic occurrences. The training and validation dataset mainly comprises normal weather conditions, which might not fully capture extreme environmental phenomena like snowfall, sandstorms, or extended cloudy days. This restricts the model's ability to generalize under such conditions. To address this, future studies should include heterogeneous datasets representing a wider variety of meteorological conditions.

Another limitation is the generalizability of the AI model to various geographic regions. The existing system is trained and tested with data gathered from Sitapura, Jaipur, Rajasthan, India. Because solar irradiance and environmental conditions change from one region to another, the same model might not offer the best performance for other geographic regions without adjustment. Transfer learning methods can be employed to adapt the model efficiently for new regions without investing in significant amounts of data collection and retraining from scratch.

Regarding scalability, the computational overhead becomes very high when the model is applied for multi-agent tracking in large-scale solar farms. The current study addresses one-to-one solar panel tracking, but when solar trackers run in parallel, the computational complexity becomes much higher. Distributed AI optimization methods need to be explored by future studies to enhance efficiency in multi-agent settings.

In addition, the delay in AI-driven decision-making is a limitation. Although edge computing minimizes delays, real-time AI inference still causes slight processing delays in adjustment response. This can affect the responsiveness of the system, especially under fast-changing environmental conditions. To address this, future deployments may use hardware-accelerated AI models (e.g., FPGA or ASIC-based AI processors) for lower latency and real-time decision-making.

Finally, the absence of blockchain incorporation for decentralized energy management is a possible limitation. AI-optimized energy generation can be further boosted by incorporating blockchain-based peer-to-peer energy trading, which allows safe sharing and exchange of surplus energy among dispersed power consumers. Although this is beyond the scope of the current research, it is an interesting area of future research.

Future work

The most promising area for future work is the creation of a hybrid AI method that unites deep learning, reinforcement learning, and bio-inspired optimization methods. Through the combination of multiple AI methods, the system will be able to attain more dynamic and adaptive solar tracking, facilitating improved performance in uncertain environmental conditions. Hybrid AI models can self-learn and adapt their tracking parameters in real time, with further optimization of solar energy conversion.

Another key enhancement is the increase in global validation with transfer learning for multi-location deployment. Rather than retraining AI models individually for various geographic regions, pre-trained AI models can be fine-tuned with little localized data to ensure scalability without huge computational costs. This method will enable easy AI adaptation to varied environmental conditions.

In addition, future research would do well to develop the intersection of AI with blockchain-based energy trading platforms. AI-based solar tracking can be supplemented with safe, decentralized, and transparent peer-to-peer energy transactions, with excess energy from one solar farm being dynamically transferred to another on a distributed ledger system. This will considerably promote smart grid optimization and energy management in solar networks.

Another enhancement is the design of AI-optimized hybrid energy storage systems. By combining lithium-ion batteries with supercapacitors, AI algorithms can dynamically switch between storage technologies depending on real-time energy supply and demand. This will reduce intermittency and stabilize the distribution of energy from solar farms.

Furthermore, a low-power edge AI-based tracking system must be engineered. Deploying AI models on FPGA or ASIC hardware can go a long way in lowering the power usage and parallelly not-keeping real-time tracking speed behind. Such solutions will have to be the focal point of solar farms that have the need for the power-efficient processing of the AI. For mass solar deployment, multi-agent reinforcement learning-based (MAREL) algorithm can be implemented by pooling the efforts of multiple solar trackers to have them work together in the most effective way as a team. The whole procedure will give solar farms a huge boost in their efficiency by distributing computational tasks and by optimizing overall energy production. Lastly, the adoption of Explainable AI (XAI) techniques should be done to improve model interpretability and transparency. Alongside SHAP (Shapley Additive Explanations), LIME (Local Interpretable Model-Agnostic Explanations), or attention-based mechanisms will show how correspondingly distillate AI models makes decisions of solar tracking that, in turn, will give rise to more trust and reliability in the energy AI.

Discussion

The findings of this study seem to support the hypothesis that solar energy tracking systems operated by AI outperform Fixed-Tilt and MPPT systems with regard to energy production, adaptability, and thermal efficiency. Using artificial intelligence based tracking, integrating machine learning, predictive analytics, and reinforcement learning, panels are adjusted to capture the maximum energy while exposing the minimum amount of energy to be lost through blades, this greatly improves the energy captured compared with previously employed systems. The power output comparison shows more energy is captured as the AI based tracking system has the ability to forecast and set the panels in the best position to capture the energy. In contrast, Fixed-Tilt systems not only restrict movement but also capture less energy along with the MPPT systems that do not go beyond adjusting the voltage. These factors allow AI powered systems the ability to forecast shifts in the direction of sunlight along with changing where the panels are directed towards in order to capture more sun, rendering the possibility of inefficient tracking obsolete. Another key finding is the impact of temperature on efficiency, and even so,

performance level heat was found to be an efficient AI based is superior to all other conventional methods such as fixed angle tilt systems. Like all methods, AI based did lose efficiency but was able to prevent more losses under harsher and more extreme conditions. This makes AI solar tracking devices an optimal solution for attitude responsiveness's.

Conclusion

This research validates that AI-based solar tracking systems are much more energy efficient compared to traditional Fixed-Tilt and MPPT tracking systems in energy efficiency, minimized thermal losses, and fast real-time response speed. With machine learning, predictive modeling, and real-time environmental monitoring, AI-based tracking systems dynamically optimize panel angles to optimize solar energy harvesting under varying environmental conditions. The findings validate that AI-based optimization delivers a remarkable energy yield improvement, as seen in the power output comparison. Compared to Fixed-Tilt systems with fixed orientations, and MPPT trackers with sole dependence on voltage control, AI-based models forecast and respond to solar movement in real-time, leading to higher energy gain. The dynamic panel angle adjustment feature enhances AI-based solar-tracking systems' efficiency in maintaining continuous sun alignment, thereby minimizing energy loss. The research also verifies the temperature impact on PV efficiency. Higher module temperatures lead to voltage drops and increased resistance and hence efficiency loss. AI-based tracking, however, offsets thermal degradation with temperature-aware optimization algorithms that dynamically manage the panel angles to prevent any heat buildup. The approach ensures consistent performance even at high-temperature operations, thus AI-based tracking is extremely effective for hot climate solar installations. Another major advantage of AI-based solar tracking is that it supports real-time response. The results show that AI-based tracking demonstrates the quickest response time (~150–200 ms), compared to MPPT (~500 ms) and Fixed-Tilt (~800 ms). The extremely fast adjustment is facilitated through edge computing, reinforcement learning, and predictive modeling, allowing instant realignment for maximum sun irradiance exposure. To provide reliability to the study, the experiment was performed on September 6, 2024, a day with uniform solar irradiance and minimal external interference. This experimental configuration ruled out environmental inconsistencies, and the resulting enhancements were solely due to AI-based tracking efficiency and not due to atmospheric irregularities. The results of this research highlight that an AI-based solar tracking system is a most advanced and efficient solution for the modern photovoltaic system. To use it in real time application it provides always real-time intelligence, the predictive analytics, and the self-learning algorithms makes AI-based tracking a highly adaptive system.

Data availability

The datasets generated and/or analysed during the current study are available in the supplementary files provided with this article.

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Author contributions

All authors have written the paper.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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